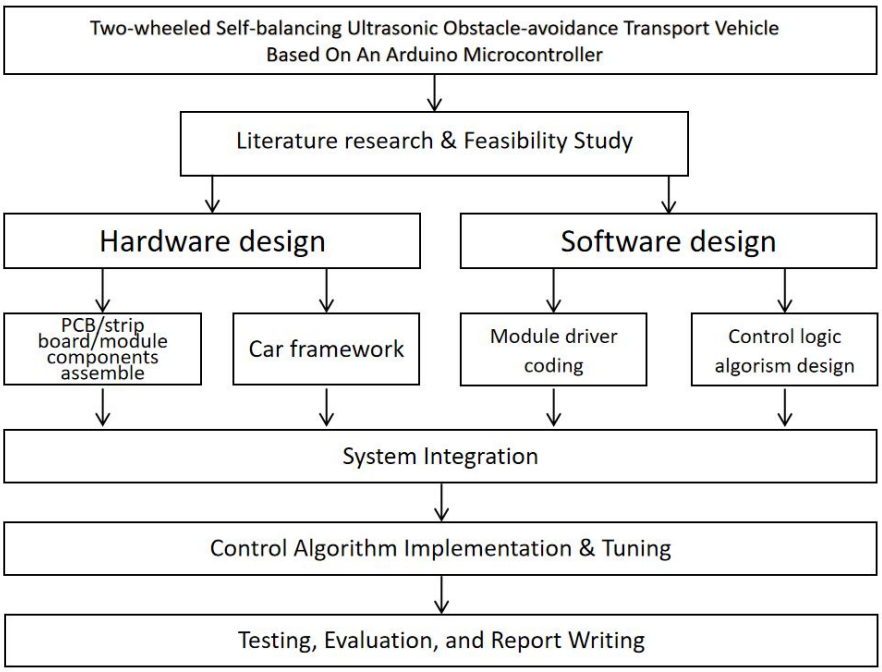
Flow-Diagram



Milestone

Month	Milestone / Activity	Description
Oct-25 - Nov-25	Modular Components coding/PID algorism design	Research self-balancing control principles, finalize hardware modules (Arduino, TB6612, MPU6050, HC-SR04, HC-05, encoders). PID algorism design
Nov(mid)-25 to Dec-25	Hardware design and assembly & individual module testing	Complete car framework, circuits wiring and verify each component: motor driver, encoder, Bluetooth, ultrasonic, MPU.
	System integration and tuing	Tune the PID controller for balance; test with collected IMU data.
Jan-26	System integration & turning algorithm	Integrate subsystems into one codebase; implement direction and movement control logic, obstacle avoidance.
Feb-26	Full-system testing & optimization	Test obstacle avoidance, balance stability, and response on different terrains; adjust parameters.
Mar-26	Final documentation & demonstration	Complete final report, prepare presentation, and demonstrate working prototype.